

$$\text{pole } x(t) = e^{st}$$

$\Downarrow$

$$x[k] = e^{s_k T}$$

$$\Downarrow$$

$$X(z) = \sum_{k=0}^{\infty} e^{s_k T} \cdot z^{-k} = \sum_0^{\infty} (e^{sT} \cdot z^{-1})^k = \frac{1}{1 - \frac{e^{sT}}{z}} :$$

$$e^{sT} = z$$

$$e^{j(\omega T + 2\pi k)} = e^{sT} = e^{jT(\omega + \frac{2\pi k}{T})}$$

limit the poles to a strip in the s-plane.

$$\text{i.e., } \left[ -\frac{\omega_s}{2}, \frac{\omega_s}{2} \right]$$